

UC Berkeley California-China Climate Institute

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Aligning markets and mandates: Insights from U.S. renewable energy credits for China's green certificates

Summary

China's green certificate program has become a cornerstone of its power-sector modernization and renewable energy governance. Since 2023, the program has linked directly to provincial targets for renewable energy consumption. However, gaps remain in attributing green certificate retirements toward provincial target compliance. This brief examines U.S. tracking frameworks for renewable energy credits that support state renewable portfolio standards, drawing lessons from institutional design and case studies. It concludes with insights on clarifying institutional roles, standardizing tracking systems, and leveraging green certificates beyond mandatory compliance to inform China's ongoing reforms toward a standardized, verified, and functionally robust certificate-based system.

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1. China's evolving green certificate program and its challenges

China's green certificate program, introduced in 2017,¹ defines each certificate as representing one megawatt-hour (MWh) of electricity produced from renewable sources such as wind, solar, hydropower, biomass, geothermal, and ocean energy.² Within a few years, the green certificate program became a key element of China's power sector modernization and was emphasized in its 14th Five-Year Plan for a Modern Energy System (2021–2025);³ it now links directly to provincial renewable energy consumption targets (provincial targets).⁴ One remaining challenge is how green certificates should be retired and credited toward fulfilling provincial targets. In this regard, U.S. experience with renewable energy credit (REC) tracking systems and integrations with state renewable portfolio standards (RPSs) may offer valuable insights for improving the design and functionality of China's green certificate program.

1.1. Evolution of China's green certificate program

Unlike in the U.S., where RECs drive the buildout of new renewable energy resources, China's green certificate program was introduced after it had already become the world's largest renewable power producer.⁵ In the U.S., state RPS programs create demand for RECs by requiring their acquisition by utilities and suppliers for compliance. In contrast, China's green certificate program began as a voluntary substitute for its feed-in tariff (FIT) system: by selling green certificates, renewable energy generators would forgo their right to FIT payments.⁶ Between 2017 and 2020, participation in China's green certificate program was minimal: of the roughly 300 million certificates issued, only about 4,300 (0.1%) were purchased, largely because no mandatory procurement obligations existed.⁷

To address this, the National Development and Reform Commission (NDRC) and the National Energy Administration (NEA) phased out premium prices for renewables under FIT scheme,⁸ mandating that all renewable projects commissioned after January 1, 2023

¹ NDRC, NEA and Ministry of Finance (2017).

² NEA (2024).

³ NDRC and NEA (2022).

⁴ NDRC, NEA and Ministry of Finance (2023).

⁵ Anders Hove and Gary Sipeng Xie (2023).

⁶ Along with the rapid decrease in renewable technology costs, the expansion of China's renewable energy deployment was largely driven by its FIT scheme. However, this created significant funding shortfalls in terms of government subsidization of renewable energy projects: because fixed tariff rates lagged behind falling renewable costs, capacity expansion soon far surpassed government-set targets. See NDRC, NEA and Ministry of Finance (2017).

⁷ Anders Hove and Gary Sipeng Xie (2023).

⁸ NDRC and NEA (2019a).

must issue certificates.⁹ At the same time, China introduced provincial targets¹⁰ specifying the share of each province's electricity consumption that must come from renewables. Provinces and covered entities—mainly retail companies and major energy consumers—can use green certificates as proof of compliance.¹¹

1.2. Challenges in linking green certificates to provincial targets

Since 2023, China has aimed to strengthen the link between green certificates and provincial targets.¹² Previously, many covered entities met provincial targets simply by consuming renewable electricity (i.e., without using certificates),¹³ but a series of recent policies increasingly emphasize certificate-based compliance.¹⁴

The intended mechanism works as follows. The NEA Business Qualification Management Center (the National Center) issues green certificates via the national registry.¹⁵ Provinces and covered entities then retire green certificates demonstrating renewable energy production, and the certificates retired in each province are counted toward fulfilling annual renewable energy consumption targets. This links verified generation data in the national registry with provincial compliance accounting.

Yet, barriers remain in attributing green certificates to provincial targets:

- Role coordination: Provincial Development and Reform Commissions (DRCs) and energy bureaus, which enforce provincial targets, lack standardized procedures to ensure that nationally recorded green certificate retirements are properly credited toward provincial target fulfillment and reconciled with any direct-consumption accounting.
- Multiple trading venues: Bundled (power + certificate) and unbundled (certificate-only) transactions occur across the national registry and regional and provincial power exchanges.¹⁶ All transactions must synchronize with the national registry, but provinces still require uniform attribution rules so retirements from any platform can be credited to their targets.

⁹ NEA (2025a).

¹⁰ NDRC and NEA (2024); NDRC and NEA. (2023); NDRC and NEA (2019b).

¹¹ NDRC and NEA (2019b); NDRC and NEA (2024).

¹² NDRC, NEA and Ministry of Finance (2023).

¹³ Anders Hove and Gary Sipeng Xie (2023).

¹⁴ NEA (2025b); NDRC, NEA, Ministry of Industry and Information Technology, Ministry of Commerce and National Data Administration (2025); NEA (2025c).

¹⁵ NEA (2024).

¹⁶ Unbundled green certificates can trade on both national registry and regional power exchanges, while in-province trade of bundled green certificates is handled at provincial centers and cross-provincial bundled transactions are traded at Beijing/Guangzhou/Inner Mongolia centers. See NEA (2024).

- **Data alignment:** Grid companies and power exchanges submit monthly dispatch and metering data to the national system, but provincial DRCs and energy bureaus need access to this data to confirm that provincial compliance accounts match national issuance and retirement records.

While China's green certificate trading framework is largely in place, these gaps point to the need for an integrated, standardized system, one with clear institutional responsibilities, verified data flows, and unified requirements for certificate-based compliance.

U.S. experience offers useful insights. State-level RPS programs depend on RECs tracked through regional registries that ensure data integrity, prevent double counting, and define institutional responsibilities. Understanding how RECs are issued, tracked, and retired, and how these processes underpin RPS compliance, can inform China's ongoing green certificate reforms.

2. Renewable energy credits and tracking systems in the U.S.

As of 2025, 29 states and Washington, D.C. have adopted mandatory RPSs or clean energy standards,¹⁷ while six states maintain voluntary goals. As the core compliance instrument verifying renewable generation, delivery, and use, each REC represents the renewable attribute of 1 MWh of renewable generation.¹⁸ To ensure integrity, 23 RPS states require REC tracking through state or regional registries, while five states with voluntary goals use tracking systems to support voluntary REC procurement (see Table 1).¹⁹

Table 1. Overview of U.S. RPS programs and tracking systems by 2025

Tracking system	Participating RPS states/territories	Launch year	Type of generation tracked	Geographic coverage	Number of generators	Number of account holders	Number of certificates issued
ERCOT	TX	2001	Renewable	Single state	420	1,236	147,337,944

¹⁷ Galen Barbose (2024).

¹⁸ Different state RPS programs define "REC" slightly differently, and the specific attributes embodied in a REC can vary. See Clean Energy States Alliance (n.d.).

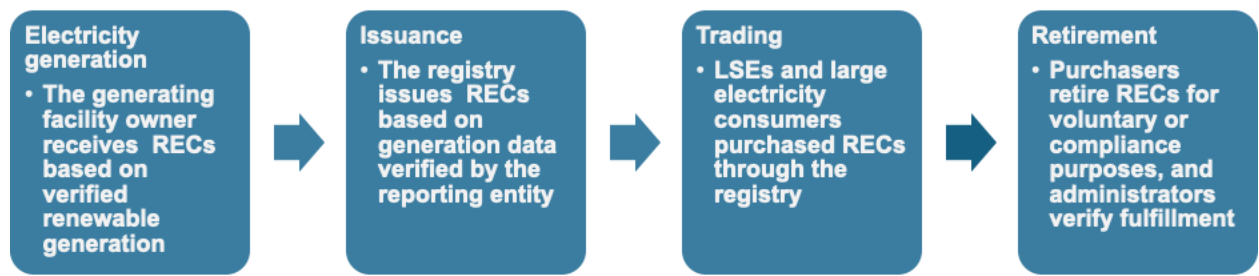
¹⁹ EPA Green Power Partnership (2025).

NEPOOL-GIS	CT, MA, ME (partial), NH, RI, VT	2002	All	Multi-state	115,632	3,234	44,609,017
PJM-GATS	DC, DE, IL (partial), MD, NJ, OH (partial), PA, VA	2005	All	Multi-state	436,583	28,100	95,316,892
M-RETS	IA, IL (partial), MN, ND, OH (partial), WI	2007	Renewable	Multi-state	3,962	684	355,738,312
WREGIS	CA, CO, NM, NV, OR, UT (starting 2025), WA	2007	Renewable	Multi-state	11,849	1,294	Not available to the public
MIRECS	MI	2009	Renewable	Single state	326	163	21,808,260
NAR	KS, ME (partial), MO, NC (out-of-state generators), NY (out-of-state generators), PR	2009	Renewable	Multi-state	947	511	264,516,285
NC-RETS	NC	2010	Renewable	Single state	1,287	668	16,039,145
NYGATS	NY	2016	All	Single state	2,924	705	129,591,042

These tracking systems, established via state legislation or regional collaboration, perform three essential functions: (a) verifying RPS compliance by tracking REC issuance and retirement; (b) monitoring ownership to prevent double counting and support voluntary

markets;²⁰ and (c) disclosing source and emission data to consumers.²¹ Thus, tracking systems offer practical alternatives to physical electricity tracking, ensuring transparent and auditable compliance records.

Figure 1 The flow of REC lifecycle



2.1. Integration into RPS compliance mechanisms

Most tracking systems were created to verify the compliance of load-serving entities (LSEs) with RPS mandates (See Figure 1). For instance, California requires all LSEs to use the Western Renewable Energy Generation Information System (WREGIS) to document renewable procurement. RECs count toward compliance only when they are retired and permanently removed from circulation, thereby preventing resale or reuse.²² Tracking systems often share a similar architecture and functionality but can offer customized information for generators, utilities, regulators, and voluntary buyers. Table 2 illustrates key participants and their roles in the REC lifecycle and associated RPS compliance.

Table 2 The role of each participant in the REC lifecycle

Types of participants	Primary roles
Generating facility owner	Registered with a unique ID number; receives RECs based on verified generation; often publicly listed
LSEs	Uses RECs to demonstrate RPS compliance
REC marketer or broker	Facilitates REC transactions and may manage REC portfolios

²⁰ EPA Green Power Partnership (2025).

²¹ EPA Green Power Partnership (2025).

²² EPA Green Power Partnership (2025).

Large electricity consumer	Purchases and retires RECs to meet their procurement goals.
State RPS administrator	Uses tracking systems for streamlined, transparent RPS compliance verification
Voluntary program administrator	Verifies that RECs meet voluntary program criteria
Data reporting entity	Submits generation data (e.g. metered output) to enable REC issuance in a given tracking systems

2.2. Data reporting and verification

In wholesale electricity markets, Independent System Operators (ISOs) and Regional Transmission Organizations (RTOs) provide monthly verified generation data, with the exception of ERCOT.²³ These grid operators balance supply and demand, controlling whether and when specific generators are dispatched and how much electricity they generate at any given time.²⁴ Given their oversight role, ISOs/RTOs submit verified monthly MWh data to tracking systems to support REC issuances. Smaller generators outside ISO/RTO jurisdiction rely on third-party verifiers or, in limited cases, self-reporting subject to validation requirements.²⁵ This robust verification scheme ensures RECs are issued only for confirmed, metered renewable generation.

2.3. Emerging trends and challenges

Initially designed as compliance tools, RECs and tracking systems are expanding in both function and scope. By 2023, voluntary markets accounted for nearly half of all REC retirements, reflecting growing corporate and consumer participation.²⁶

Hourly and location-based RECs: Interest is growing in developing time-matched and location-specific RECs to better reflect real-time renewable generation and decarbonization impacts. For large electricity consumers, for example, this approach goes beyond total MWh procurements to match electricity use with renewables produced at a desired time and place (since a large share of solar generation occurs at midday, which can

²³ Federal Energy Regulatory Commission (n.d.).

²⁴ EPA Green Power Partnership (2025).

²⁵ Typically, self-reporting is allowed only for small facilities (<1 MW), subject to specific circumstances. See CRS (2023).

²⁶ EPA Green Power Partnership (2025).

still lead to fossil-based ramping in the evening).²⁷ Time-matched RECs could therefore incentivize energy-storage deployment and/or cleaner generation portfolios.²⁸

Several regional tracking systems have begun piloting or implementing this functionality. PJM-GATS introduced hourly tracking in 2023, when M-RETS and NAR completed hourly pilots with certain customers.²⁹ However, cost remains a challenge, as most single-state systems are often financed through ratepayers rather than direct user fees.³⁰ The upgrading of tracking systems will increase the bill rates by end-users. Another challenge is integrating RPS programs with relevant regulatory agencies, a process that remains undefined and will require explicit state regulatory approval. Finally, limitations around data reporting complicate hourly load matching, as much of the load served within a tracking system's footprint lies outside ISO/RTO jurisdiction.

GHG accounting and corporate reporting: The GHG Protocol provides globally recognized methods for quantifying greenhouse gas emissions.³¹ Energy Attribute Certificates, such as RECs, are used to verify renewable energy procurement and demonstrate emissions of consumed electricity. The GHG Protocol is updating its Scope 2 Guidance,³² which pertains to electricity consumption, to clarify how RECs can be used for indirect emissions reporting.³³ Beyond this guidance, corporations are increasingly being required, either voluntarily or through emerging regulatory disclosure rules, to report verified carbon emission data. In all cases, RECs continue to underpin corporate claims of clean energy procurement and Scope 2 emission reductions.

Low embodied emission products: Beyond electricity markets, RECs are becoming relevant for product lifecycle carbon accounting. Under initiatives like Buy Clean,³⁴ RECs and tracking systems support life-cycle carbon accounting, linking renewable electricity

²⁷ Chalendar and Benson (2019).

²⁸ Ben Gerber (2021).

²⁹ CRS (2023).

³⁰ CRS (2023).

³¹ In the 1990s, the World Resources Institute and World Business Council for Sustainable Development developed a set of international standards and guidance, referred to broadly as the GHG Protocol. The GHG Protocol is the most widely used standard by organizations seeking to measure their emissions in order to manage them. See GHG Protocol (n.d.a).

³² GHG Protocol (n.d.b).

³³ Purchases of unbundled RECs are not encouraged for inclusion in GHG inventory accounting. RECs purchased through a power purchase agreement or utility green power program may be included, however. See GHG Protocol (n.d.b).

³⁴ This initiative promotes construction materials with lower embodied emissions for use in public procurement and publicly-funded projects. See Federal Buy Clean Initiative (n.d.).

inputs used in production to Environmental Product Declarations³⁵ for construction materials. Such verified REC data enhance the credibility of low-carbon procurements and sustainability reporting.

3. Case studies: Tracking system governance structures in California and PJM states

The California/WREGIS case illustrates how a single state with stringent RPS requirements can leverage a tracking system to ensure rigorous, rule-based compliance. California's RPS relies on WREGIS to track, verify, and retire RECs from certified generators (see Figure 2). Renewable facilities must first obtain RPS certification from the California Energy Commission (CEC), which verifies eligibility based on site location, fuel type, metering accuracy, and compliance with the RPS Eligibility Guidebook.³⁶ Once certified, each facility receives a unique RPS ID recognized by WREGIS.³⁷ The California Independent System Operator (CAISO) provides verified generation data for grid-connected facilities,³⁸ which WREGIS uses to issue RECs on a monthly basis. LSEs then retire these RECs to demonstrate RPS compliance. RPS oversight is shared between the CEC and the California Public Utilities Commission (CPUC). The CEC verifies REC data and publishes annual compliance reports, while the CPUC reviews procurement results and enforces compliance, including penalties for shortfalls.³⁹ Together, they ensure the accuracy, transparency, and integrity of California's RPS program.⁴⁰

³⁵ Environmental product declarations summarize the results of life cycle assessment studies for specific products or product categories. See Product Category Rules (n.d.).

³⁶ California Energy Commission (n.d.).

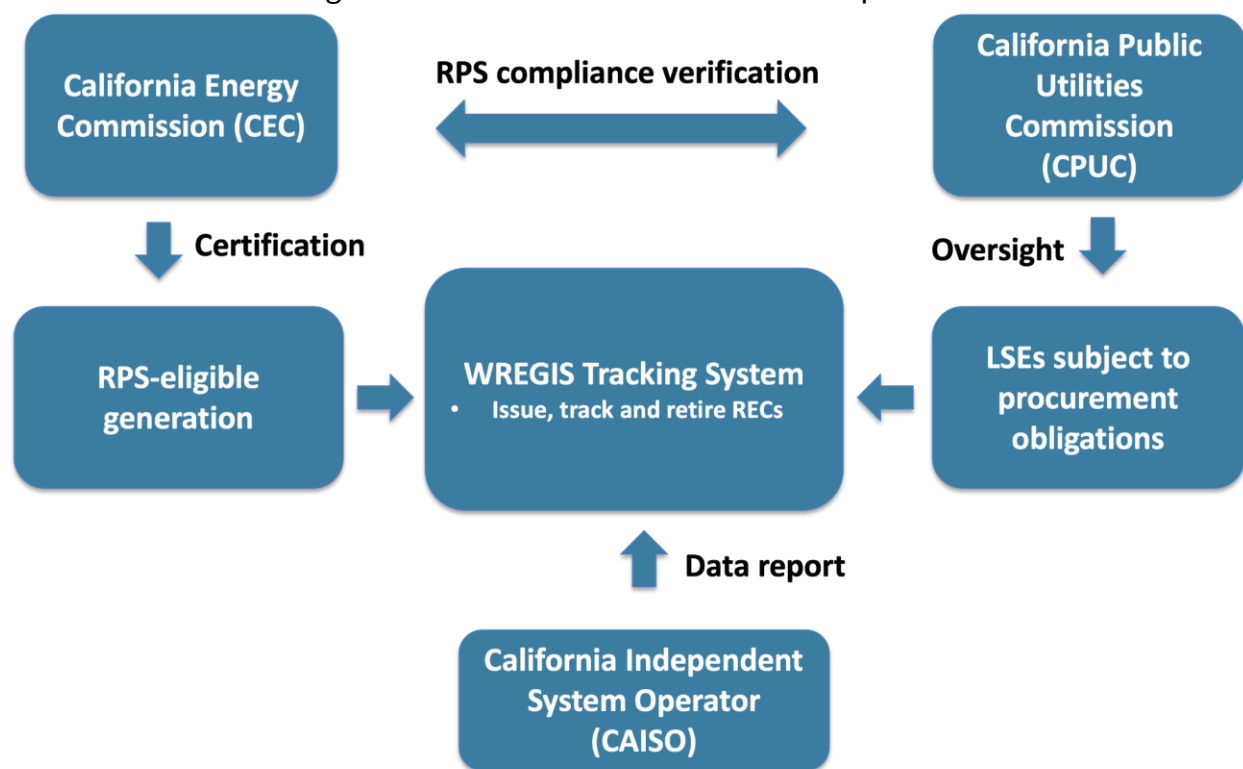
³⁷ California Energy Commission (n.d.).

³⁸ California ISO (2016).

³⁹ California Energy Commission (n.d.b).

⁴⁰ California Energy Commission (n.d.b).

Figure 2 The flow of California RPS compliance



By contrast, the PJM states/PJM-GATS example shows, from a regional perspective, how a shared tracking system can support multiple state RPS programs under a more flexible and market-oriented model. PJM-GATS is a multi-state platform used across the Mid-Atlantic region⁴¹ to support state RPS compliance. State public utility commissions (PUCs) certify RPS-eligible generators and assign state-specific certification numbers. Based on each state's RPS targets, several entities - LSEs, utilities, and retail suppliers - must procure and retire RECs equivalent to a set share of their retail electricity sales. Unlike California's in-state model, PJM-GATS states generally accept RECs generated in other PJM jurisdictions for compliance, subject to state-specific eligibility rules. State PUCs use data supported by the RTO, PJM Interconnection,⁴² to verify whether entities have met their annual RPS targets and technology carve-outs.⁴³ In cases of non-compliance, obligations may be fulfilled by making alternative compliance payments.⁴⁴

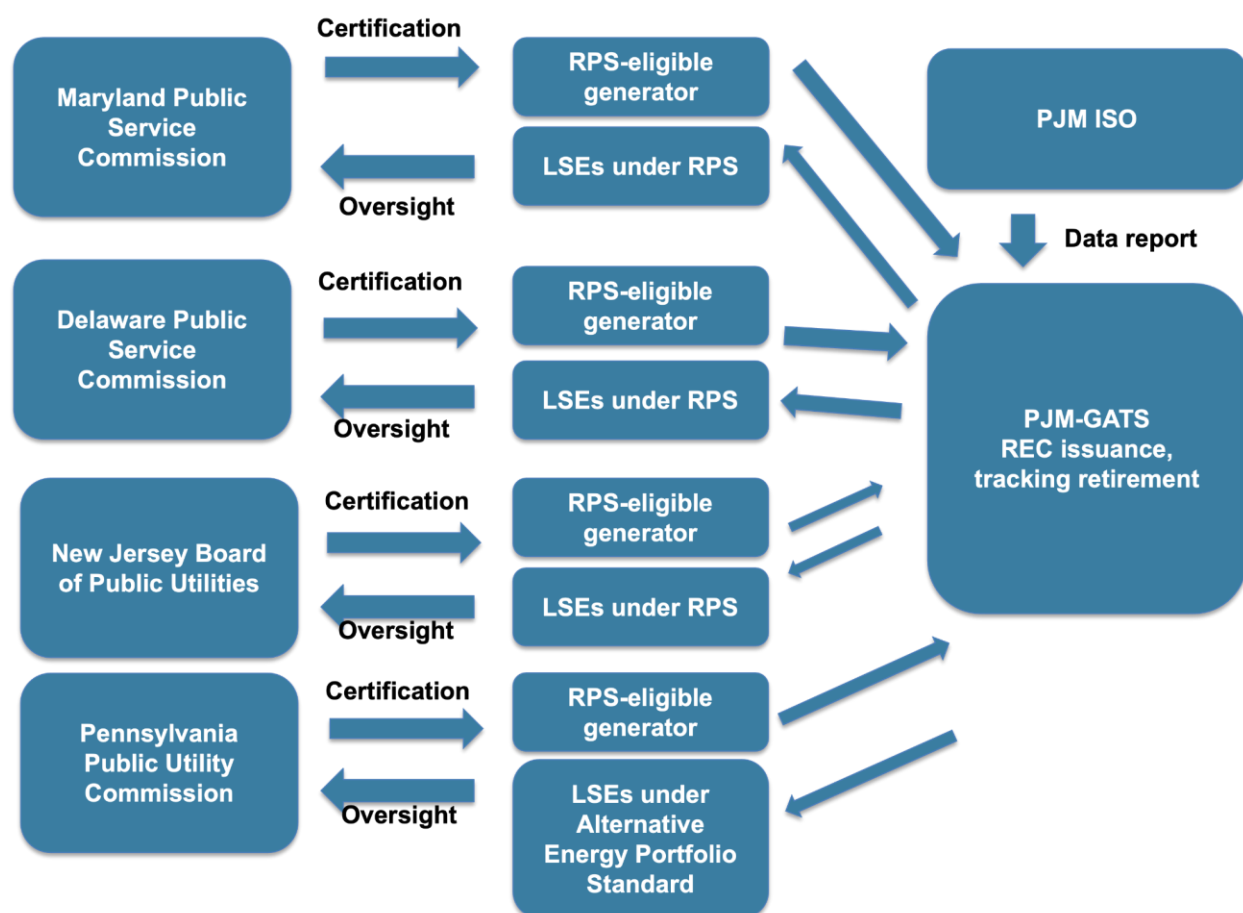
⁴¹ This region includes New Jersey, Maryland, Pennsylvania, Delaware, and Washington, D.C. See Maryland Renewable Energy Portfolio Standard (n.d.); Delaware Renewables Portfolio Standard (n.d.); New Jersey Renewables Portfolio Standard (n.d.); Pennsylvania Alternative Energy Portfolio Standard (n.d.).

⁴² PJM Interconnection (named for founding states Pennsylvania, New Jersey, and Maryland) is an RTO serving all or parts of 13 U.S. states and the District of Columbia. See PJM (n.d.).

⁴³ Carve-out policies set aside a portion of a state's energy goals to help advance certain types of renewable technologies. See Climate XChange (2022).

⁴⁴ U.S. EPA (2022).

Figure 3 The flow of RPS compliance in PJM states



While both California and PJM states utilize centralized tracking systems to integrate RECs into their respective RPS compliance frameworks, their approaches differ in regulatory design. REC use in California is tightly governed by the CEC and CPUC. Only bundled RECs (i.e., those contractually linked to electricity delivered in California or generated by CEC-certified facilities) count towards RPS compliance. This framework intentionally limits flexibility to localize economic and environmental benefits. Although WREGIS itself is policy-neutral, California's own strict eligibility and delivery criteria demonstrate how a state can shape a regional tracking platform to meet its specific regulatory goals.

In contrast, PJM-GATS provides a uniform platform for issuing, transferring, and retiring RECs across participating Mid-Atlantic states, with each state's PUC setting its own RPS eligibility rules. Most PJM states permit unbundled RECs, including those imported from other PJM jurisdictions, and allow cross-state trading subject to state-specific limits. This design maximizes efficiency and cost-effectiveness, accommodating diverse state mandates and voluntary REC trading within a unified regional framework.

4. Conclusion: Lessons for China

In the U.S., RECs—supported by state and regional tracking systems—are the foundation for verifying RPS compliance. State PUCs and ISOs/RTOs have well-defined roles in both verification and data reporting, ensuring that only eligible generation is certified and that REC retirements are properly recorded for compliance. The California/WREGIS model prioritizes stringent eligibility criteria and in-state benefits within a tightly regulated framework, whereas for PJM states, PJM-GATS provides trading efficiency and cross-state trading flexibility on a shared regional platform that accommodates various state policies. As China expands its green certificate program and links it more closely to provincial targets, several institutional lessons from the U.S. can inform improvements in policy design and market performance.

4.1. Clarifying institutional roles

U.S. programs delineate clear responsibilities among state regulators, grid operators, tracking systems, and market participants. This improves accountability throughout the REC lifecycle—from certification, issuance, and trading to retirement. For China, clarifying the roles of key actors is essential. As illustrated in Table 3, for instance, the national registry, operated by the National Center, can manage certificate issuance, transfer, and retirement at the national level. Provincial DRCs and energy bureaus can set and enforce provincial targets, determine eligible generation, review compliance filings, and coordinate enforcement. Grid companies and power exchanges can submit verified metered generation and delivery data, while independent third-party verifiers may supplement oversight for distributed or non-dispatchable resources.

Table 3 Comparing institutional roles in attributing certificates to provincial targets

Functionality	U.S. responsible entity	Potential Chinese responsible entity
Certificate issuance, trading, and retirement	Regional/Statewide tracking system, (e.g. WREGIS, PJM-GATS)	National center
Setting and enforcing mandatory renewable consumption targets	State RPS administrators (e.g. CPUC and CAISO, and state PUCs in PJM states)	Provincial DRCs and energy bureaus

Data verification	RTO/ISOs (e.g. CAISO, PJM-ISO)	Grid companies and power exchange centers
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4.2. Developing standardized tracking platforms across provinces

U.S. tracking systems function as centralized, policy-neutral registries that ensure integrity from REC creation to retirement. As illustrated by the California/WREGIS model, a stringent regulatory framework can ensure compliance with state-level renewable targets by localizing interests, whereas the PJM states/PJM-GATS model provides greater flexibility, enabling efficient mechanisms for cross-state trading and unified market operation.

Therefore, China could draw on insights from both models. Its national registration system could adopt a similar design concept—maintaining a unified national framework while granting provinces a certain degree of flexibility to accommodate regional differences, such as delivery requirements or eligibility vintages. Such a design would support cross-provincial trading where appropriate, while maintaining a unified system architecture and consistent data standards.

4.3. Leveraging green certificates beyond today's mandates

In the U.S., RECs are also used for voluntary corporate procurement and reporting, as well as emerging time- and location-based matching practices. Similarly, China can extend the use of green certificates to support broader decarbonization and life-cycle accounting efforts.⁴⁵ While these new functionalities hold promise, challenges remain in clarifying recognition rules for voluntary claims, enhancing data accuracy, and aligning regulatory oversight. Ultimately, a credible and flexible green certificate program—supported by clear institutional roles, a policy-neutral national registry that accommodates provincial rules, and verified metered data—will be essential for aligning China's renewable energy accounting with its broader decarbonization goals.

⁴⁵ NDRC, NEA and Ministry of Finance (2023).

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About the California-China Climate Institute

The California-China Climate Institute was launched in September 2019 and is a University of California-wide initiative, housed jointly at UC Berkeley's School of Law (through its Center for Law, Energy, and the Environment) and the Raussier College of Natural Resources. It is Chaired by Jerry Brown, former Governor of the State of California, and Vice-Chaired by the former Chair of the California Air Resources Board, Mary Nichols. The Institute also works closely with other University of California campuses, departments, and leaders. Through joint research, training, and dialogue in and between California and China, this Institute aims to inform policy makers, foster cooperation and partnership and drive climate solutions at all levels.